

 Edit



Integrate Azure AI Immersive Reader into Your Application

Challenge Overview

Understand the scenario

You are a Developer for Hexelo, an organization that needs to integrate accessibility tools into their own applications, enhancing the user experience.

In this Challenge Lab, you will integrate AI Immersive Reader into your Application. First, you will set up the Azure AI Immersive Reader service and configure your application, and then you will translate input text into multiple languages by using Azure AI Immersive Reader. Next, you will highlight parts of speech within text, and then you will break words into syllables to enhance readability. Finally, you will provide visual aids by displaying images to improve understanding.

Navigating the Challenge Lab

🕒 ► Understanding a Cloud Slice

🕒 ► Quick tips for navigating the Challenge Lab instructions.

[New to Challenge Labs? Click here to learn more.](#)

Set up the Azure AI Immersive Reader service and configure your application

Hints Enabled

No Yes

Note About Evolving Technology

Azure AI technologies are constantly evolving as Microsoft continues to update, refine, and improve its tools and interfaces. As a result, you may notice slight differences between the instructions in this lab and what you see on your screen. Rest assured, we are actively working to keep all tutorials up to date with the latest features and interfaces. We appreciate your patience and understanding, and we apologize for any inconvenience this may cause.

- Sign into the virtual machine as **Admin** using `T` `Passw0rd!` as the password.
- Open **Microsoft Edge**, go to `T` `https://portal.azure.com`, sign in to the Microsoft Azure portal as `T` `{USERNAME}` using `T` `{PASSWORD}` as the password, and then dismiss all prompts.

Expand this hint for guidance on signing into the Microsoft Azure portal.

- On the Windows taskbar, in Search, enter **Edge**, and then select **Microsoft Edge** to open a new browser.
- In the Address bar, enter `T` `https://portal.azure.com`, and then select **Enter**.
- In the Sign in dialog, enter `T` `{USERNAME}`, and then select **Next**.
- In Enter password, enter `T` `{PASSWORD}` and then select **Sign in**.
- Dismiss all prompts.

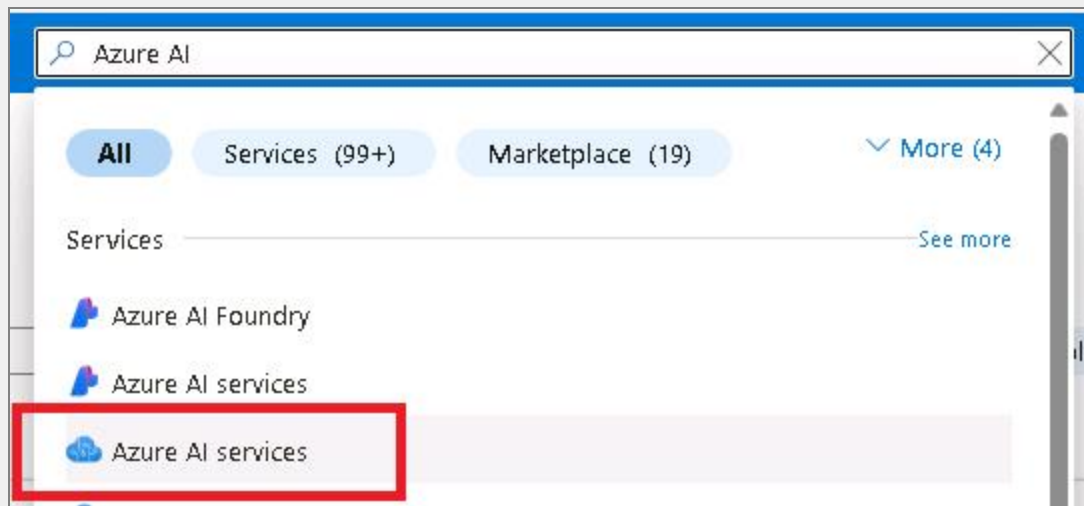
 Want to learn more? Review the documentation on [the Microsoft Azure portal](#).

- Create an Azure AI Immersive Reader resource named `T` `hexelo-immersive{LAB_INSTANCE_ID}` by using the values in the following table. For any property that is not specified, use the default value.

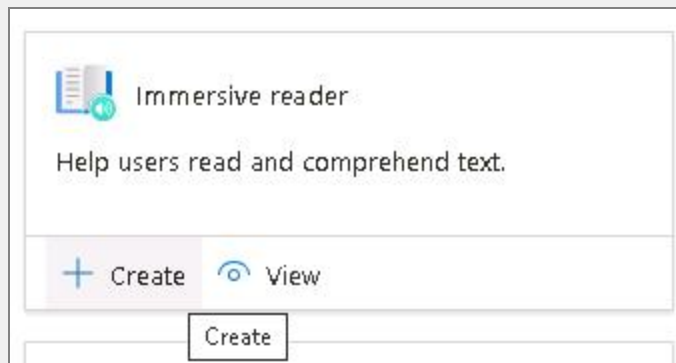
Property	Value
Resource group	{RESOURCE_GROUP_NAME}
Region	East US
Name	T hexelo-immersive{LAB_INSTANCE_ID}
Pricing Tier	Commercial S0

💡 Expand this hint for guidance on creating an Azure AI Immersive Reader resource. ^

- On the Microsoft Azure portal navigation bar, in *Search resources, services, and docs*, search for and select **Azure AI Services**.



- On the Azure AI services blade, on the Overview page, in **Immersive reader**, select **Create**.



- On the Create Immersive Reader blade, on the Basics tab, in Project details, in Resource group, select **{RESOURCE_GROUP_NAME}**.
- In Instance Details, in Region, select **East US**.
- In Name, enter **hexelo-immersive{LAB_INSTANCE_ID}**.
- In Pricing tier, select **Commercial S0 (600 Sessions per minute)**, select **Review + create**, and then select **Create**.

Create Immersive Reader

Basics Network Identity Tags Review + create

Embed text reading and comprehension capabilities into applications. Help users of any age and reading ability with features like reading aloud, translating languages, and focusing attention through highlighting and other design elements. Azure is the only major cloud provider offering this type of reading technology. No machine learning expertise is required.

[Learn more](#)

Project Details

Subscription * ⓘ Skillable Original Content Dev 01

Resource group * ⓘ RG1lod50142151

[Create new](#)

i Azure AI services resource creation requires subscription registration, we detected that your selected subscription did not register Cognitive services resource type before, we will help you to register Cognitive services resource type when you select a subscription in subscription dropdown. Click to learn more how to check registration state for your selected subscription.

Instance Details

Region ⓘ East US

Name * ⓘ hexelo-immersive50142151 ✓

Pricing tier * ⓘ Commercial S0 (600 Sessions per minute)

[View full pricing details](#)

Previous

Next

Review + create

 Want to learn more? Review the documentation on [creating an Azure AI Immersive Reader resource](#).

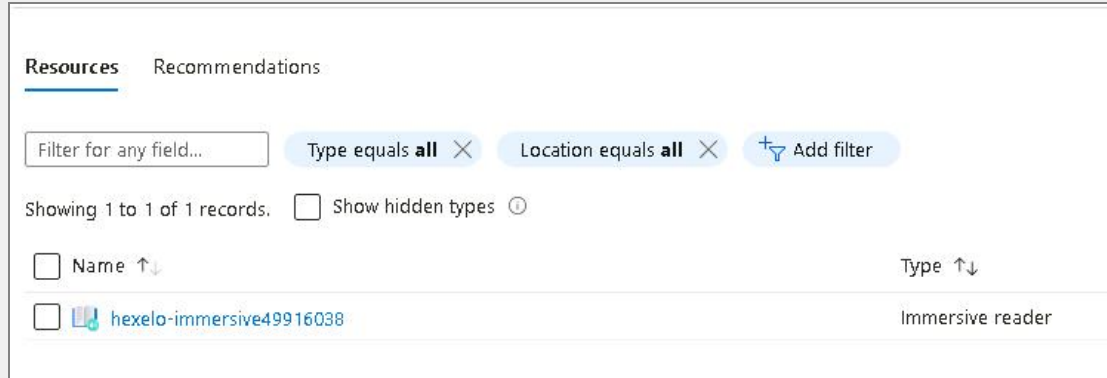
- Locate the hexelo-immersive{LAB_INSTANCE_ID} **API Key** and **Endpoint URL**, copy the values, and then paste the values into the following **Immersive Reader API Key** and **Endpoint URL** text boxes.

Immersive Reader API Key

Endpoint URL

💡 Expand this hint for guidance on obtaining the API Keys and Endpoint URL. ^

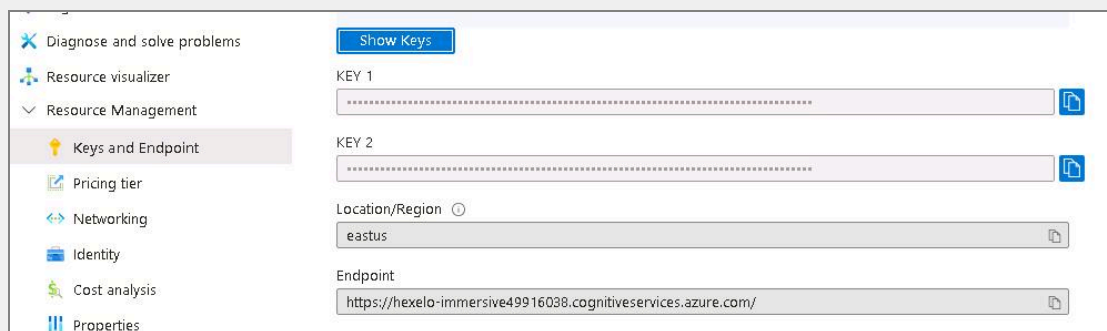
- On the Deployment blade, select **Go to resource**.
- On the {RESOURCE_GROUP_NAME} blade, in Resources, select **hexelo-immersive{LAB_INSTANCE_ID}**.



- On the hexelo-immersive{LAB_INSTANCE_ID} service menu, in Resource Management, select **Keys and Endpoint**.



- On the Keys and Endpoint page, select **Show Keys**, select the **Copy to Clipboard** button to the right of KEY 1, and then paste the value into the **API Key** text box in the Challenge Lab instructions.



- Select the **Copy to Clipboard** button to the right of Endpoint, and then paste the value into the **Endpoint URL** text box in the Challenge Lab instructions.

💡 Want to learn more? Review the documentation on [obtaining the API Keys and Endpoint URL](#).

- Open **T** Command Prompt as an administrator, and then install the Azure AI Immersive Reader SDK by running the following commands:

Bash

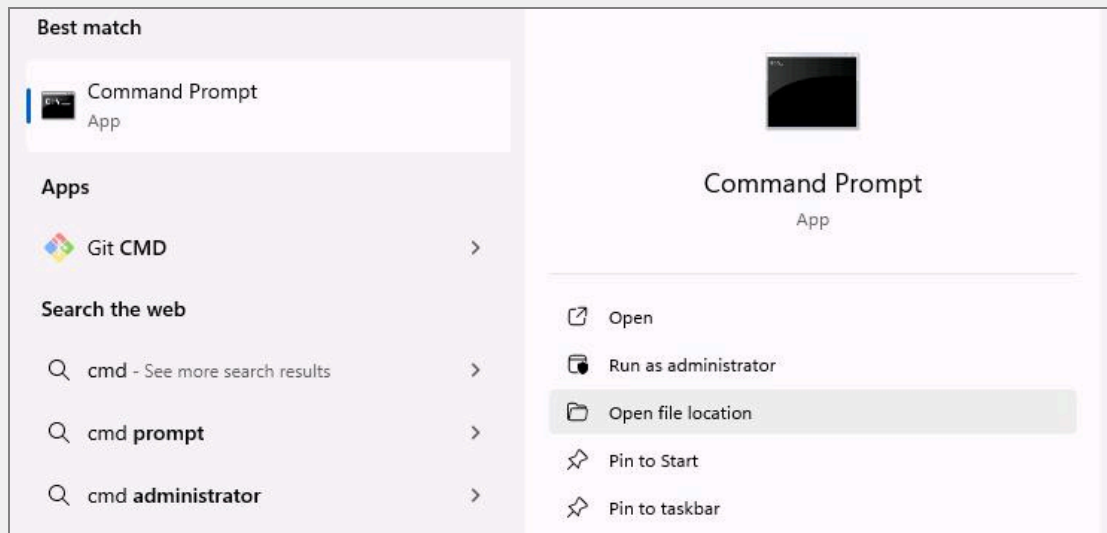
```
npm i @microsoft/immersive-reader-sdk
```

Bash

```
npm install @azure/ms-rest-js
```

💡 Expand this hint for guidance on installing Node.js.

- On the Windows taskbar, in Search, enter **T** CMD, and then in Command Prompt, select **Run as administrator**.



- In the User Account Control dialog, select **Yes**.
- Enter the following commands to install Node.js.

Bash

```
npm i @microsoft/immersive-reader-sdk
```

Bash

```
npm install @azure/ms-rest-js
```



Want to learn more? Review the documentation on [installing Node.js](#).

- Set the environment variables by running the following commands:

Bash



```
set IMMERSIVE_READER_KEY==<APIKey>
```



```
set IMMERSIVE_READER_ENDPOINT=<EndpointURL>
```



Want to learn more? Review the documentation on [setting environment variables](#).



Minimize **Command Prompt** for later use. Leave **Microsoft Edge** open.

Check your work

Verify

Translate input text into multiple languages by using Azure AI Immersive Reader

Hints Enabled

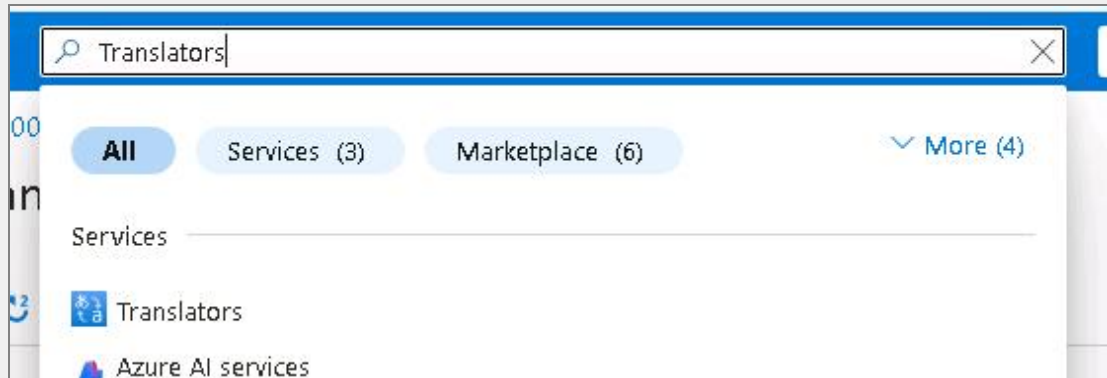
No Yes

- Create a new T Azure AI Translator service named T TranslationService{LAB_INSTANCE_ID} by using the values in the following table. For any property that is not specified, use the default value.

Property	Value
Resource group	{RESOURCE_GROUP_NAME}
Region	East US
Name	T TranslationService{LAB_INSTANCE_ID}
Pricing Tier	Standard S1

💡 Expand this hint for guidance on creating an Azure AI Translator service.

- On the Microsoft Azure portal navigation bar, in *Search resources, services, and docs* (G+), search for and select **T** Translators.



- On the Azure AI services blade, on the Translator page, select **+ Create Translator**.



- On the Create Translator blade, on the Basics tab, in Project Details, in Resource group, select **{RESOURCE_GROUP_NAME}**.
- In Instance Details, in Region, select **East US**, and then in Name, enter **T** TranslationService{LAB_INSTANCE_ID}.
- In Pricing Tier, select **Standard S1 (Pay as you go)**, select **Review + create**, and then select **Create**.

Create Translator

Project Details

Subscription * ⓘ

Skillable Original Content Dev 01

Resource group * ⓘ

RG1lod49916038

Create new

ⓘ

Azure AI services resource creation requires subscription registration, we detected that your selected subscription did not register Cognitive services resource type before, we will help you to register Cognitive services resource type when you select a subscription in subscription dropdown. Click to learn more how to check registration state for your selected subscription.

Instance Details

ⓘ

Please choose the Global region unless your business or application requires a specific region. Applications that do not offer a region selection use the Global region.

Region * ⓘ

East US

Name * ⓘ

TranslationService49916038

Pricing tier * ⓘ

Standard S1 (Pay as you go)

[View full pricing details](#)

 Want to learn more? Review the documentation on [creating an Azure AI Translator service](#).

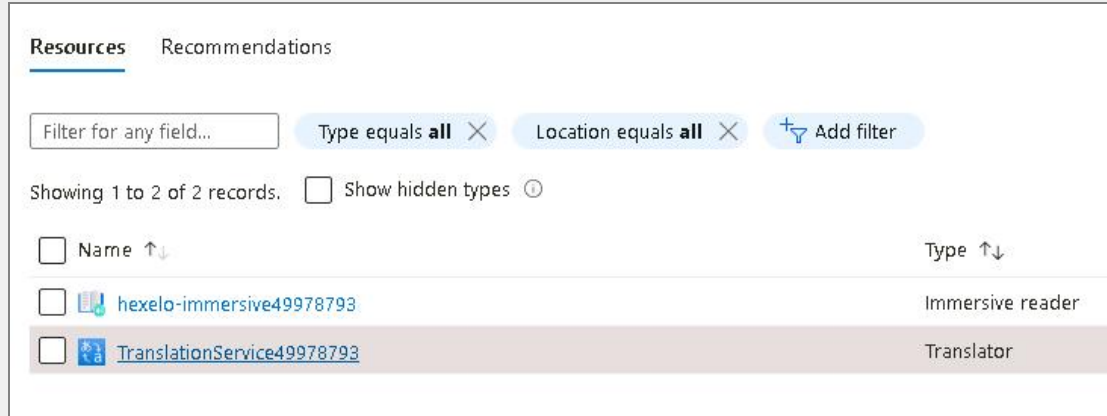
- Locate the TranslationService{LAB_INSTANCE_ID} **API Key** and **Web API Text Translation Endpoint**, copy the values, and then paste them into the following **Translator API Key** and **Web API Text Translation Endpoint URL** text boxes:

Translator API key

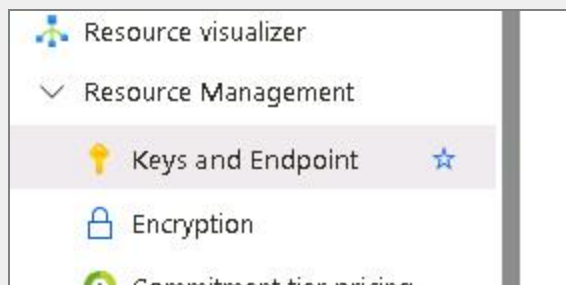
Web API Text Translation Endpoint URL

💡 Expand this hint for guidance on locating the API Key and Endpoint.

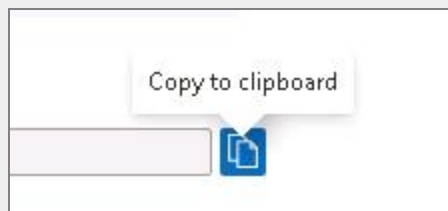
- On the Deployment blade, select **Go to resource**.
- On the {RESOURCE_GROUP_NAME} blade, on the Overview page, in Resources, select **TranslationService{LAB_INSTANCE_ID}**-the link, not the checkbox.



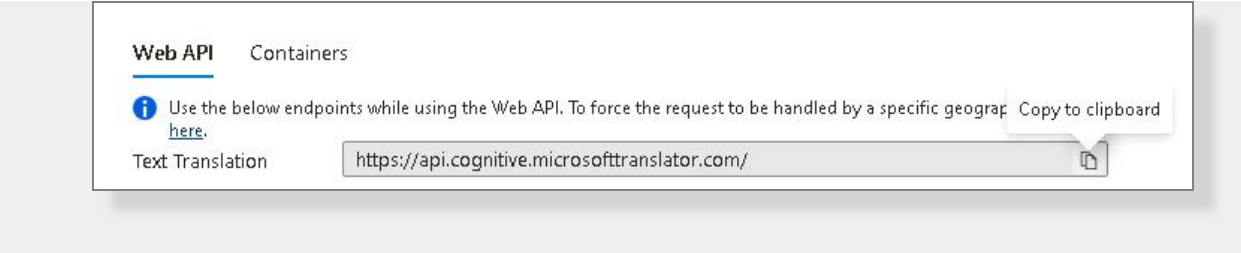
- On the TranslationService{LAB_INSTANCE_ID} service menu, in Resource Management, select **Keys and Endpoint**.



- On the Keys and Endpoint page, select the copy-to-clipboard button to the right of **Key 1**, and then paste the copied value into the **API Key** text box in the Challenge Lab instructions.



- In Web API, select the copy-to-clipboard icon to the right of **Text Translation**, and then paste the copied value into the **Web API Text Translation Endpoint URL** text box in the Challenge Lab instructions.



- Open a new browser tab, go to `[T]` https://github.com/LODSContent/ChallengeLabs_Resources/blob/master/AI102/translatorindex.html, download the **TranslatorIndex.html** file, and then move it to the `[T]` `D:\LabFiles` folder.

 Expand this hint for guidance on downloading documents. 

- Open a new **Microsoft Edge** browser tab, go to `[T]` https://github.com/LODSContent/ChallengeLabs_Resources/blob/master/AI102/translatorindex.html and then on the `AI102/translatorindex.html` page, select the **Download raw file** button.
- In the Downloads dialog, if needed, select **Keep**, select the **Show in folder** button and then move the file from the Downloads folder to the `[T]` `D:\LabFiles` folder.

- Open the `D:\LabFiles\translatorindex.html` file by using **Notepad**, and then replace the original values by using the new values as displayed in the following table:

Original Value	New Value
<code>[T]</code> <code>YOUR_TRANSLATOR_KEY</code>	<code>[T]</code> <code><TranAPIKey></code>
<code>[T]</code> <code>YOUR_TRANSLATOR_REGION</code>	<code>[T]</code> <code>EastUS</code>
<code>[T]</code> <code>YOUR_IMMERSIVE_READER_KEY</code>	<code>[T]</code> <code><APIKey></code>
<code>[T]</code> <code>YOUR_RESOURCE_NAME</code>	<code>[T]</code> <code>TranslationService{LAB_INSTANCE_ID}</code>

💡 Expand this hint for guidance on editing the html file. ^

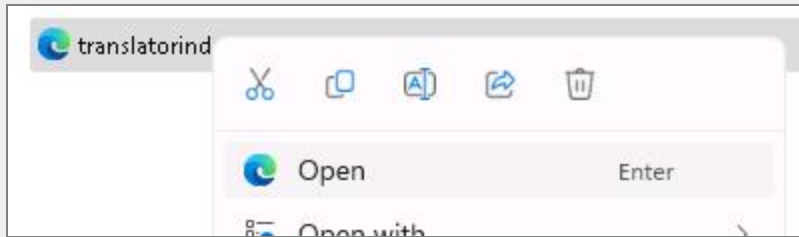
- Open **Explorer**, go to **D:\LabFiles**, right-click **translatorindex.html**, select **Open with**, select **Choose another app**, in More options select **Notepad**, and then select **Just once** .
- In Notepad, search for `[T] YOUR_TRANSLATOR_KEY` and then replace it with `[T] <TranAPIKey>`.
- Search for `[T] YOUR_TRANSLATOR_REGION` and then replace it with `[T] EastUS`.
- Search for `[T] YOUR_IMMERSIVE_READER_KEY` and then replace it with `[T] <APIKey>`.
- Search for `[T] YOUR_RESOURCE_NAME` and then replace it with `[T] TranslationService{LAB_INSTANCE_ID}`.
- Save the file.

- Open the **D:\LabFiles\translatorindex.html** file by using **Microsoft Edge** , and then translate the following sentence to **Spanish**:

`[T]` I just created an artificial intelligence web application that will translate text for me.

💡 Expand this hint for guidance on opening an html file in a web browser. ^

- In **File Explorer**, right-click **D:\LabFiles\translator.html**, and then select **Open**.



- On the Azure AI Immersive Reader page, in the text translation textbox, enter the following: **T** I just created an artificial intelligence web application that will translate text for me.
- In Choose Language, select **Spanish**, and then select **Translate**.

Azure AI Immersive Reader - Text Translation

I just created an artificial intelligence web application that will translate text for me.

Choose Language: Spanish ▼

Translate Open in Immersive Reader

Translated Text: Acabo de crear una aplicación web de inteligencia artificial que traducirá texto por mí.

- Translate the above sentence to **German**, and then enter the translation into the following **German Translation** text box:

German Translation

Check your work

Verify

Highlight parts of speech, split syllables, and provide visual aids by using Azure AI Immersive Reader

Hints Enabled

No Yes

- Create an HTML file named `HighlightIndex.html` in the `D:\LabFiles` folder that highlights each line of text as you select it in yellow, by using the following html code:

 In this Challenge Lab, use the **Copy** feature rather than the Type Text feature- this will avoid formatting issues. 

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Azure AI Immersive Reader</title>
  <style>
    .highlight {
      background-color: yellow;
    }
  </style>
</head>
<body>
  <div id="content">
    <p id="line1">Azure AI Immersive Reader is a powerful tool.</p>
    <p id="line2">It helps with reading comprehension and accessibility.</p>
    <p id="line3">Selecting each line with your mouse will highlight that l
  </div>

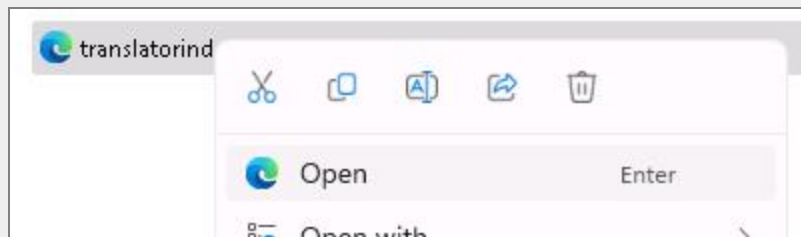
  <script>
    document.querySelectorAll('#content p').forEach(line => {
      line.addEventListener('click', () => {
        line.classList.toggle('highlight');
      });
    });
  </script>
```

```
</body>  
</html>
```

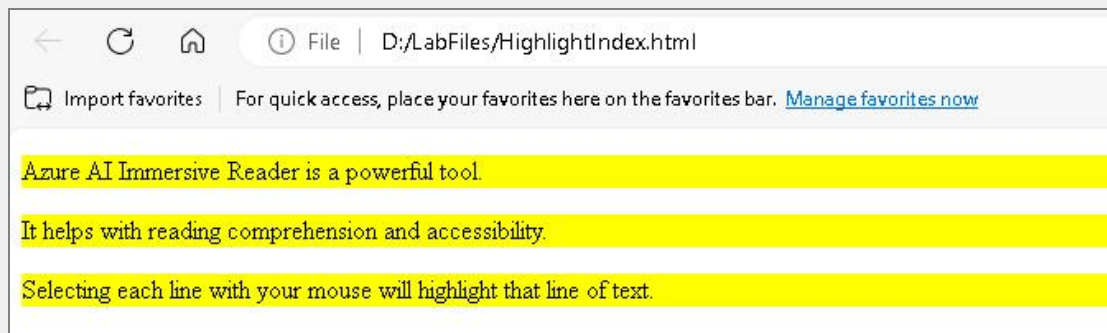
- Open **D:\LabFiles\HighlightIndex.html** in a new **Microsoft Edge** browser tab, select each line of text to confirm that it becomes highlighted, and then select each line again to remove the highlight.

💡 Expand this hint for guidance on opening an html file.

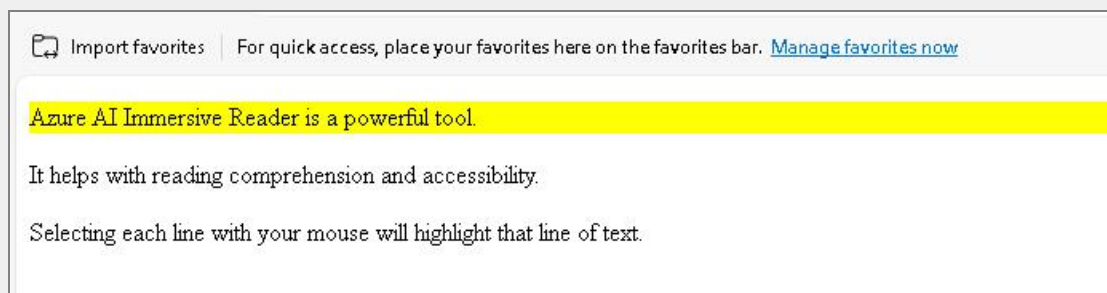
- Right-click **D:\LabFiles\HighlightIndex.html** and select **Open**.



- On the Azure AI Immersive Reader page, select each line of text.



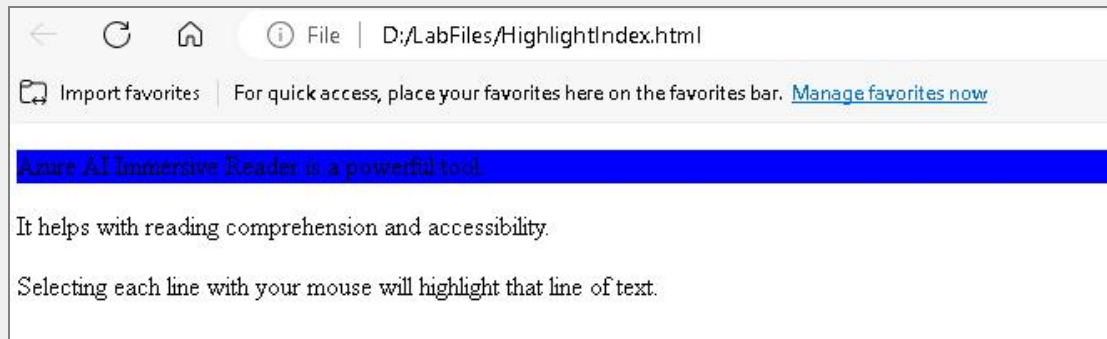
- Select each line again to remove the highlight from that line of text.



- Modify the **D:\LabFiles\HighlightIndex.html** file by using **Notepad** to change the highlight color to **blue**, open the file by using **Microsoft Edge**, and then select a line to confirm that the highlight color is now **blue**.

💡 Expand this hint for guidance on editing an html file.

- Right-click **D:\LabFiles\HighlightIndex.html**, and then select **Open with | Notepad**.
- On line 9, change the color **yellow** to **blue**.
- Select **File | Save**, and then open the file in a new **Microsoft Edge** web browser tab.
- Select a line to highlight it and then confirm that the highlight color is blue.



- Create a new file to break words into syllables by using **Notepad** and the following HTML code, and then save it to **D:\LabFiles\SyllableIndex.html**:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Syllable Splitter</title>
  <style>
    .highlight {
      background-color: yellow;
    }
    .syllable {
      border-bottom: 2px dashed;
    }
  </style>
</head>
<body>
  <div id="content">
    <p id="line1">Azure AI Immersive Reader is a powerful tool.</p>
    <p id="line2">It helps with reading comprehension and accessibility.</p>
  </div>

  <div id="text">
    Breaking words into syllables helps with readability.
  </div>

  <script>
```

```

document.querySelectorAll('#content p').forEach(line => {
    line.addEventListener('click', () => {
        line.classList.toggle('highlight');
    });
});

const textElement = document.getElementById('text');
const textContent = textElement.innerText;
const colors = ['red', 'blue', 'green', 'purple', 'orange'];

function splitIntoSyllables(word) {
    let colorIndex = 0;
    return word.replace(/([aeiouy]+[^\aeiouy]*)/g, (match) => {
        const color = colors[colorIndex % colors.length];
        colorIndex++;
        return `<span class="syllable" style="border-bottom-color: ${color}`
    });
}

textElement.innerHTML = textContent.split(' ').map(splitIntoSyllables).join(' ');
</script>
</body>
</html>

```

- Open **D:\LabFiles\SyllableIndex.html** by using **Microsoft Edge** and review the output.
- Create a new file that will display visual aids by using **Notepad** and the following HTML code, and then save it to **D:\LabFiles\VisualIndex.html**:

```

<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Visual Aid Integration</title>
    <style>
        #visual-aid img {
            width: 100px;
            margin: 5px;
        }
    </style>
</head>
<body>
    <div id="text">
        The cat sat beside the hat on the mat and the rat ran behind the bat.
    </div>
    <div id="visual-aid"></div>

    <script>

```

```

document.addEventListener("DOMContentLoaded", function () {
    const imageMap = {
        'cat': 'https://postimg.cc/kRFk8z88',
        'bat': 'https://postimg.cc/5Yvc962s',
        'rat': 'https://postimg.cc/7C58cB88',
        'hat': 'https://postimg.cc/d7gPK8w7',
        'mat': 'https://postimg.cc/D4K9SnWn'
    };

    const textContent = document.getElementById('text').innerText.split(' ');
    const visualAidDiv = document.getElementById('visual-aid');

    textContent.forEach(word => {
        const lowerWord = word.toLowerCase().replace(/^[a-z]/g, ''); // Remove first letter
        if (imageMap[lowerWord]) {
            const img = document.createElement('img');
            img.src = imageMap[lowerWord];
            img.alt = lowerWord;
            img.onerror = function () {
                this.src = 'https://i.imgur.com/placeholder.png'; // Fallback
            };
            visualAidDiv.appendChild(img);
        }
    });
});
</script>
</body>
</html>

```

- Open **D:\LabFiles\VisualIndex** by using **Microsoft Edge** and review the output.

Check your work

Verify

Summary

Congratulations, you have completed the **Integrate Azure AI Immersive Reader Into Your Application** Challenge Lab.

You have accomplished the following:

- Set up the Azure AI Immersive Reader service and configured your application.
- Translated input text into multiple languages by using Azure AI Immersive Reader.
- Highlighted parts of speech within text by using Azure AI Immersive Reader.
- Broke words into syllables to enhance readability.
- Provided visual aids by displaying images to improve understanding.

Ending your lab

To ensure your lab is recorded as complete, select **Submit** or **End** below. Exiting [X] the lab will result in an incomplete status.

 Once you select **Submit** or **End**, you will not be able to return to this Challenge Lab.

Your feedback is important!

As you end your Challenge Lab, please take a few minutes to complete the short survey that will appear in the next window. Alternatively, you may provide your feedback directly to [Challenge Labs feedback](#).

Looking for your next Challenge Lab?

Below are recommended Challenge Labs to try next.

- Explore Document Processing in Azure AI's Document Intelligence Studio [Guided]
- Create a Bot by Using the Bot Framework SDK [Guided]
- Generate Insights from Images by Using Azure AI Vision Services [Guided]
- Build an AI Model in Azure AI Machine Learning Studio [Guided]
- Integrate Azure AI Immersive Reader Into Your Application by Using Its SDK [Guided]

- Summarize Text Automatically by Using Azure AI Language Services [Guided]
- Translate Text Between Languages by Using Azure AI Translator [Guided]
- Generate Image Descriptions from Text by Using Azure AI Services [Guided]
- Build an AI-Powered Search System by Using Azure AI Search [Guided]
- Analyze and Generate Text-Based Sentiment by Using Azure AI Services [Guided]

